

● MEDIUM

● INFORMATION AND IDEAS

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03

10 questions · ~15 min

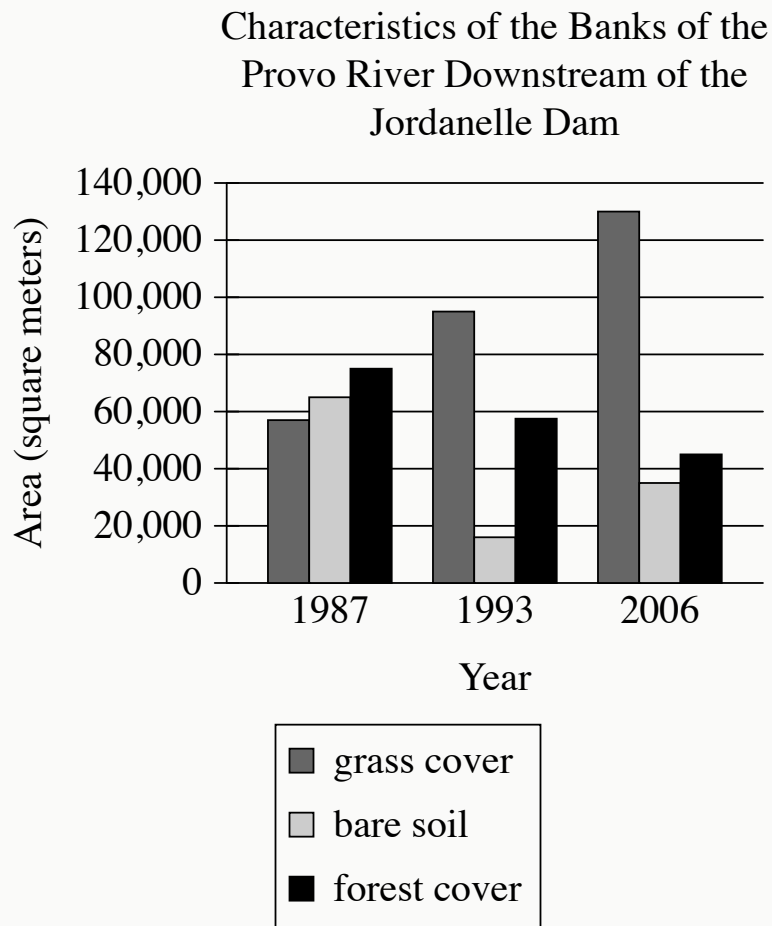
Q1

INFORMATION AND IDEAS

Herbivorous sauropod dinosaurs could grow more than 100 feet long and weigh up to 80 tons, and some researchers have attributed the evolution of sauropods to such massive sizes to increased plant production resulting from high levels of atmospheric carbon dioxide during the Mesozoic era. However, there is no evidence of significant spikes in carbon dioxide levels coinciding with relevant periods in sauropod evolution, such as when the first large sauropods appeared, when several sauropod lineages underwent further evolution toward gigantism, or when sauropods reached their maximum known sizes, suggesting that _____blank

Which choice most logically completes the text?

- A. fluctuations in atmospheric carbon dioxide affected different sauropod lineages differently.
- B. the evolution of larger body sizes in sauropods did not depend on increased atmospheric carbon dioxide.
- C. atmospheric carbon dioxide was higher when the largest known sauropods lived than it was when the first sauropods appeared.
- D. sauropods probably would not have evolved to such immense sizes if atmospheric carbon dioxide had been even slightly higher.



- For each data category, the following bars are shown:
 - grass cover
 - bare soil
 - forest cover
- All values are approximate.
- The data for the 3 categories are as follows:
 - 1987:
 - grass cover: 57,000 square meters
 - bare soil: 65,000 square meters
 - forest cover: 75,000 square meters
 - 1993:
 - grass cover: 95,000 square meters

- bare soil: 16,000 square meters
- forest cover: 57,500 square meters
- 2006:
 - grass cover: 130,000 square meters
 - bare soil: 35,000 square meters
 - forest cover: 45,000 square meters

The Jordanelle Dam was built on the Provo River in Utah in 1992. Earth scientist Adriana E. Martinez and colleagues tracked changes to the environment on the banks of the river downstream of the dam, including how much grass and forest cover were present. They concluded that the dam changed the flow of the river in ways that benefited grass plants but didn't benefit trees.

Which choice best describes data from the graph that support Martinez and colleagues' conclusion?

- A. The lowest amount of grass cover was approximately 58,000 square meters, and the highest amount of forest cover was approximately 75,000 square meters.
- B. There was more grass cover than forest cover in 1987, and this difference increased dramatically in 1993 and again in 2006.
- C. There was less grass cover than bare soil in 1987 but more grass cover than bare soil in 1993 and 2006, whereas there was more forest cover than bare soil in all three years.
- D. Grass cover increased from 1987 to 1993 and from 1993 to 2006, whereas forest cover decreased in those periods.

A subject of much speculation, distinctive sets of parallel ridges mark the icy crust of Europa, Jupiter's smallest moon. Researchers now claim that the ridges' formation mechanism mirrors that of a strikingly similar pair on Greenland's ice sheet. There, surface water seeped through fissures in the sheet and formed a water pocket that subsequently disrupted the overlying ice, forcing fragments of it upward and outward into peaks, as the pocket froze and expanded. Although Europa lacks liquid surface water, the same process could be driven by the moon's subsurface ocean.

Which choice best states the main idea of the text?

- A. Researchers think that the ridges on Europa and the ridges in Greenland may have been formed by the same process even though Europa, unlike Greenland, doesn't have liquid water on its surface.
- B. The primary difference between the ridges on Europa and the ridges in Greenland is that unlike the Europa ridges, the Greenland ridges are parallel.
- C. The pair of ridges found on Greenland's ice sheet appear to have formed long before the recently discovered sets of ridges on Europa formed.
- D. Researchers don't understand why Europa is marked by so many sets of ridges when the moon doesn't have any liquid water on its surface that could have collected and expanded under the icy crust.

Red velvet cake has been a favorite dessert of many for years, but the recipe's origins are unclear. A bakery in Dallas, Texas, argued that it created the first recipe for the cake when the bakery opened in the 1860s. The Adams Extract Co., which sells baking products, claims to have created the recipe in the 1930s to help market their red dye. A US hotel and a Canadian department store also publicly stated that the red velvet cake sold in each of their establishments in the 1930s was an original creation, each alleging that it was the recipe author. No clear evidence has emerged to favor one of these claims over the others, however. It thus seems that _____ blank

Which choice most logically completes the text?

- A. red velvet cake was first baked sometime before the 1860s.
- B. we cannot say at present who actually baked the first red velvet cake.
- C. none of the supposed inventors of red velvet cake are likely to have invented it.
- D. the bakery in Dallas, Texas, probably invented red velvet cake.

“Often Rebuked, Yet Always Back Returning” is an 1846 poem by Emily Brontë. The poem conveys the speaker’s determination to experience the countryside around her: _____ blank

Which quotation from the poem most effectively illustrates the claim?

- A. “Often rebuked, yet always back returning / To those first feelings that were born with me, / And leaving busy chase of wealth and learning / For idle dreams of things which cannot be.”
- B. “I’ll walk, but not in old heroic traces, / And not in paths of high morality, / And not among the half-distinguished faces, / The clouded forms of long-past history.”
- C. “I’ll walk where my own nature would be leading: / It vexes me to choose another guide: / Where the grey flocks in ferny glens are feeding; / Where the wild wind blows on the mountain side.”
- D. “To-day, I will seek not the shadowy region; / Its unsustaining vastness waxes drear; / And visions rising, legion after legion, / Bring the unreal world too strangely near.”

A common assumption among art historians is that the invention of photography in the mid-nineteenth century displaced the painted portrait in the public consciousness. The diminishing popularity of the portrait miniature, which coincided with the rise of photography, seems to support this claim. However, photography's impact on the portrait miniature may be overstated. Although records from art exhibitions in the Netherlands from 1820 to 1892 show a decrease in the number of both full-sized and miniature portraits submitted, this trend was established before the invention of photography.

Based on the text, what can be concluded about the diminishing popularity of the portrait miniature in the nineteenth century?

- A. Factors other than the rise of photography may be more directly responsible for the portrait miniature's decline.
- B. Although portrait miniatures became less common than photographs, they were widely regarded as having more artistic merit.
- C. The popularity of the portrait miniature likely persisted for longer than art historians have assumed.
- D. As demand for portrait miniatures decreased, portrait artists likely shifted their creative focus to photography.

As the name suggests, dramaturges originated in theater, where they continue to serve a variety of functions: conducting historical research for directors, compiling character biographies for actors, and perhaps most importantly, helping writers of plays and musicals to hone the works' stories and characters. Performance scholar Susan Manning observes that many choreographers, like playwrights and musical theater writers, are concerned with storytelling and characterization. In fact, some choreographers describe the dances they create as expressions of narrative through movement; it is therefore unsurprising that _____ blank

Which choice most logically completes the text?

- A. dramaturges can have a profound impact on the artistic direction of plays and musicals.
- B. choreographers developing dances with narrative elements frequently engage dramaturges to assist in refining those elements.
- C. dances by choreographers who incorporate narrative elements are more accessible to audiences than dances by choreographers who do not.
- D. some directors and actors rely too heavily on dramaturges to complete certain research tasks.

Participants' Evaluation of the Likelihood That Robots Can Work Effectively in Different Occupations

Occupation	Somewhat or very unlikely (%)	Neutral (%)	Somewhat or very likely (%)
television news anchor	24	9	67
teacher	37	16	47
firefighter	62	9	30
surgeon	74	9	16
tour guide	10	8	82

Rows in table may not add up to 100 due to rounding.

Georgia Tech roboticists De'Aira Bryant and Ayanna Howard, along with ethicist Jason Borenstein, were interested in people's perceptions of robots' competence. They recruited participants and asked them how likely they think it is that a robot could do the work required in various occupations. Participants' evaluations varied widely depending on which occupation was being considered; for example, _____blank

Which choice most effectively uses data from the table to complete the example?

- A. 47% of participants believe that it is somewhat or very likely that a robot could work effectively as a teacher, but 37% of respondents believe that it is somewhat or very unlikely that a robot could do so.
- B. 9% of participants were neutral about whether a robot could work effectively as a television news anchor, which is the same percent of participants who were

neutral when asked about a robot working as a surgeon.

- C. 82% of participants believe that it is somewhat or very likely that a robot could work effectively as a tour guide, but only 16% believe that it is somewhat or very likely that a robot could work as a surgeon.
- D. 62% of participants believe that it is somewhat or very unlikely that a robot could work effectively as a firefighter.

At over a thousand pages across two volumes, *The Fifty-Year Mission*, compiled by Edward Gross and Mark A. Altman, is presented as the “complete, uncensored, unauthorized oral history” as told by the people behind the media franchise *Star Trek*. The work aspires to be comprehensive by, for example, including accounts from cast and crew members of every *Star Trek* television series and film to date. But while *The Fifty-Year Mission* is clearly a unique and valuable resource, it has a shortcoming common among oral histories: it lacks a clear authorial point of view that could otherwise unite the various accounts into a cohesive whole.

Which choice best states the main idea of the text?

- A. The compilers of *The Fifty-Year Mission* had lofty goals for their oral history of the *Star Trek* franchise, but the published work lacks information about many key events in the franchise’s history.
- B. *The Fifty-Year Mission* includes more accounts from people involved with *Star Trek* television shows than it does from people involved with *Star Trek* films.
- C. The large amount of material compiled into *The Fifty-Year Mission* is surprising given that many of the people involved in the *Star Trek* franchise did not participate in the oral history project.
- D. *The Fifty-Year Mission* represents a worthwhile attempt to thoroughly recount the history of the *Star Trek* franchise, but its approach has an important limitation.

Although military veterans make up a small proportion of the total population of the United States, they occupy a significantly higher proportion of the jobs in the civilian government. One possible explanation for this disproportionate representation is that military service familiarizes people with certain organizational structures that are also reflected in the civilian government bureaucracy, and this familiarity thus _____ blank

Which choice most logically completes the text?

- A. makes civilian government jobs especially appealing to military veterans.
- B. alters the typical relationship between military service and subsequent career preferences.
- C. encourages nonveterans applying for civilian government jobs to consider military service instead.
- D. increases the number of civilian government jobs that require some amount of military experience to perform.